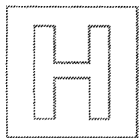


NAME: _____

CLASS: _____

INDEX: _____



CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS
Higher 2

BIOLOGY

Paper 1 Multiple Choice

9648/01

WEDNESDAY 4 SEPT 2013

1 hour 15 minutes

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write and/or shade your name, NRIC / FIN number and HT group on the Answer Sheet in the spaces provided unless this has been done for you.

There are forty questions on this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **2B pencil** on the separate Answer Sheet.

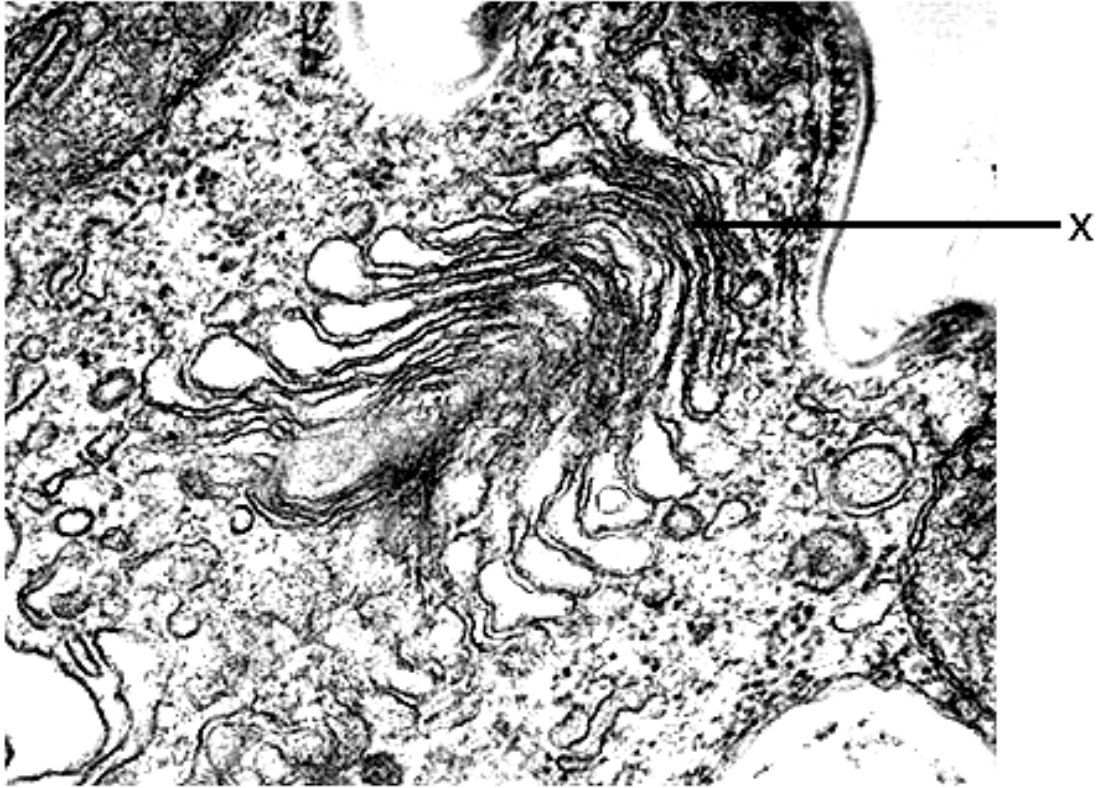
Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

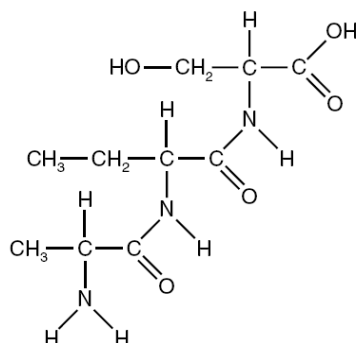
Calculators may be used.

- 1 Which of the following are the most likely consequences for a cell lacking Structure X?



- I The cell dies because it is unable to make glycoproteins to detect stimuli from its environment.
 - II The cell dies from a lack of enzymes to digest food taken in by endocytosis.
 - III The cell dies from the accumulation of worn out organelles within itself.
 - IV The cell is unable to reproduce itself.
 - V The cell is unable to export its enzymes or peptide hormones.
-
- A II and III
 - B II and V
 - C III, IV and V
 - D I, II, III and V

- 2 The diagram below shows a biological molecule found in the living cells.



The following are some statements concerning the molecule shown above.

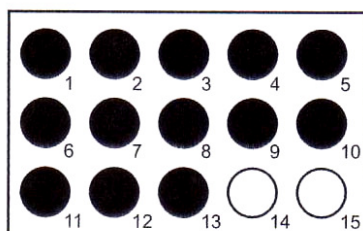
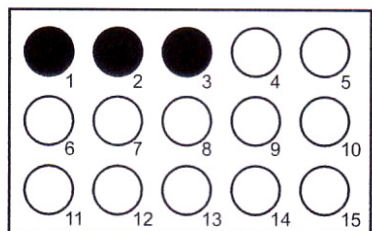
- 1 It is soluble in water.
- 2 Double bonds between the carbon and oxygen atoms increase membrane fluidity.
- 3 It can act as a buffer against extreme pH change.
- 4 It can undergo condensation to form a polymer.

Which of the above statements are true about this molecule?

- A** 1 and 3
B 1 and 4
C 2 and 3
D 2 and 4

- 3 An experiment was carried out to investigate the digestion of starch using amylase at two different temperatures. A sample was removed from each mixture at 15 second intervals and place onto a spotting tile well containing two drops of iodine in KI solution. The results are shown in the diagram.

Which shows the correct temperatures and times for the complete digestion of starch?



key



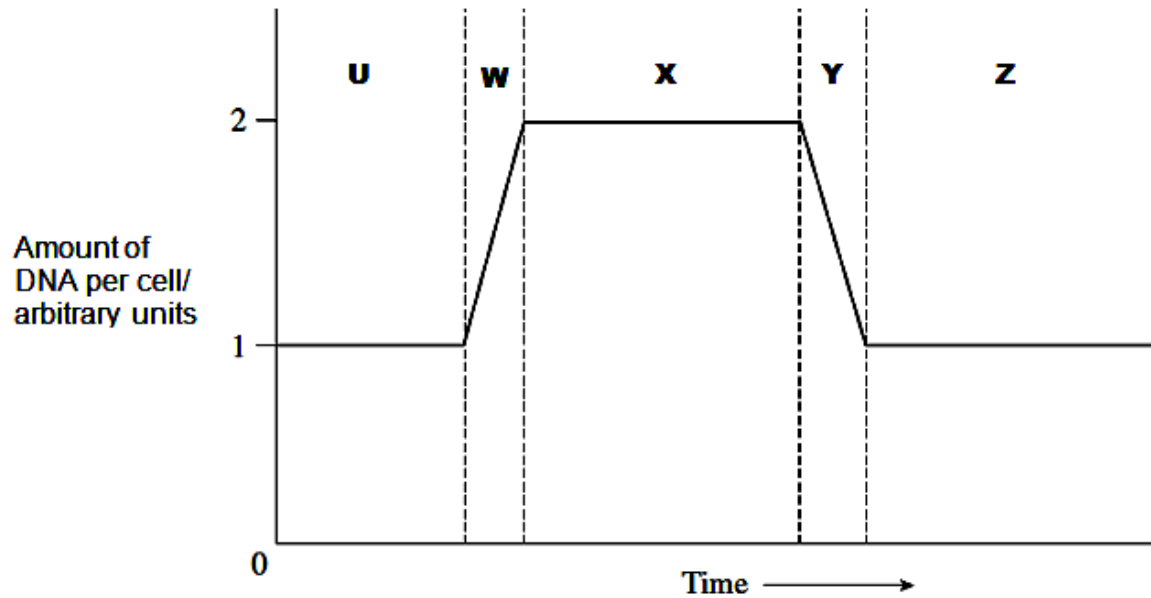
blue black



brown yellow

	temperature / °C	time / s
A	30	3.15
	10	0.45
B	30	45
	10	195
C	10	45
	30	195
D	10	3.15
	30	0.45

- 4 The graph shows changes in the amount of DNA in a cell during one cell cycle. The letters U – Z marks out the different phases in the cell cycle.



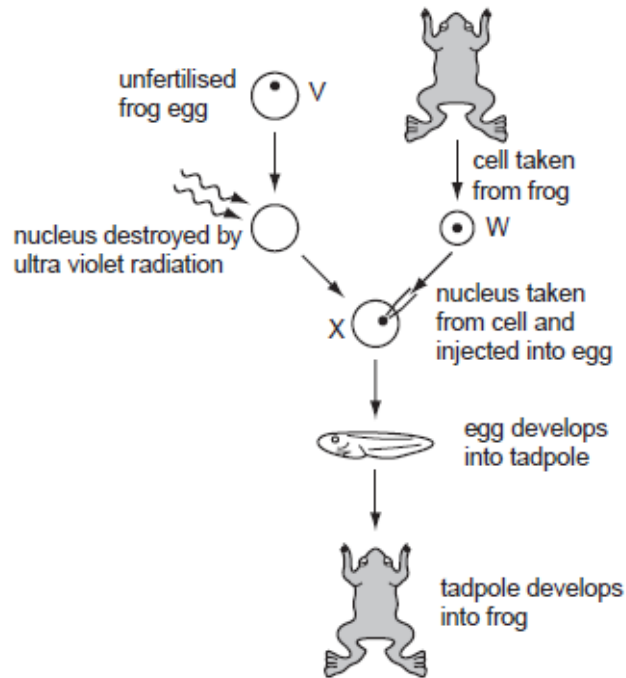
Many drugs that are used to treat cancer work at different time periods during the cell cycle.

- (i) Cisplatin binds to DNA and stops free DNA nucleotides from joining together.
- (ii) Drug B stops spindle fibres from shortening.

With reference to the cell cycle above, determine where these 2 drugs work.

	Cisplatin	Drug B
A	W	X
B	W	Y
C	U	X
D	U	Z

- 5 The diagram shows how genetically identical frogs can be developed from unfertilised frog eggs. The diploid number for frogs is 26.



Which combination of numbers correctly identifies the number of chromosomes in each type of cell?

	V	W	X
A	13	13	26
B	13	26	13
C	13	26	26
D	26	26	13

- 6 The table below shows a list of characteristics displayed by mutant strains of *E.coli* during DNA replication and the possible reasons.

No	Characteristics of mutant <i>E.coli</i>	Enzymes or functions affected by mutation
1	Okazaki fragments accumulate and DNA synthesis is never completed	DNA ligase activity is missing
2	Supercoils are found to remain at the flanks of the replication bubbles	DNA helicase is hyperactive
3	Synthesis is very slow.	DNA polymerase keeps dissociating from the DNA and has to re-associate
4	No initiation of replication occurs.	A-T rich region at origin of replication deleted

Which of the reasons correctly explain the characteristics displayed by the mutant *E. coli* strains?

- A 2 and 3
 B 1 and 3
 C 1, 3, and 4
 D All the above

- 7 Part of the amino acid sequence of protein X in normal and disease patients are shown below.

Protein X in normal patient

...Lys-His-Pro-Asp-Thr...

Protein X in disease patient

...Lys-His-Pro-Gly-Thr...

mRNA codons for the amino acids shown in the sequence are as follows:

His: CAU, CAC

Asp: GAU, GAC

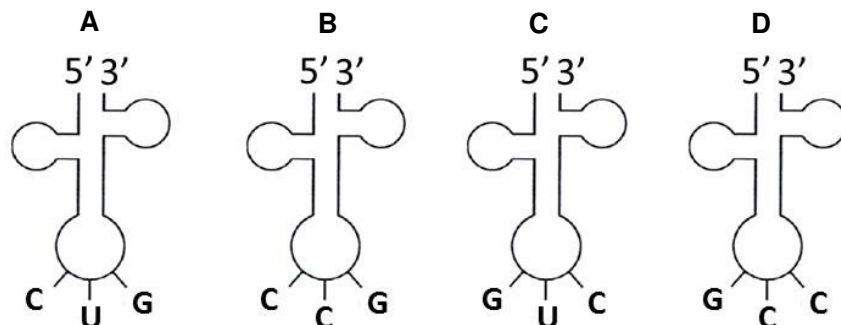
Lys: AAA, AAG

Thr: ACU, ACC, ACA, ACG

Gly: GGU, GGC, GGA, GGG

Pro: CCU, CCC, CCA, CCG

Which transfer RNA is involved in forming the modified part of protein X?



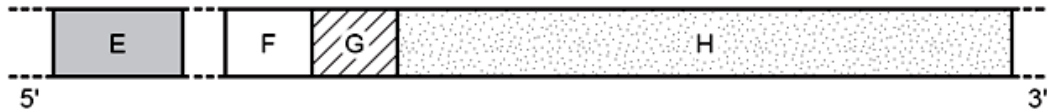
- 8 Listed below are some antibacterial drugs that can affect the synthesis of bacterial proteins.

Anti-microbial drug	Rifampicin	Streptomycin	Tetracycline
Mode of action	Binds to RNA polymerase	mRNA misread during translation	Prevents binding of tRNA to ribosome

Which row shows the correct effects of these drugs on bacterial protein synthesis?

	Rifampicin	Streptomycin	Tetracycline
A	Defective protein synthesised	mRNA does not bind to ribosome	Polymerization of amino acids does not occur
B	mRNA not synthesised	Defective protein synthesised	Polymerization of amino acids does not occur
C	mRNA not synthesised	mRNA does not bind to ribosome	Transcription prevented
D	Transcription prevented	Defective protein synthesised	mRNA does not bind to ribosome

- 9 The diagram represents a length of DNA which includes an operon. E, F, G and H have different functions.



What correctly identifies E, F, G and H?

	E	F	G	H
A	operator	structural gene(s)	regulatory gene	promoter
B	promoter	regulatory gene	structural gene(s)	operator
C	regulatory gene	promoter	operator	structural gene(s)
D	structural gene(s)	operator	promoter	regulatory gene

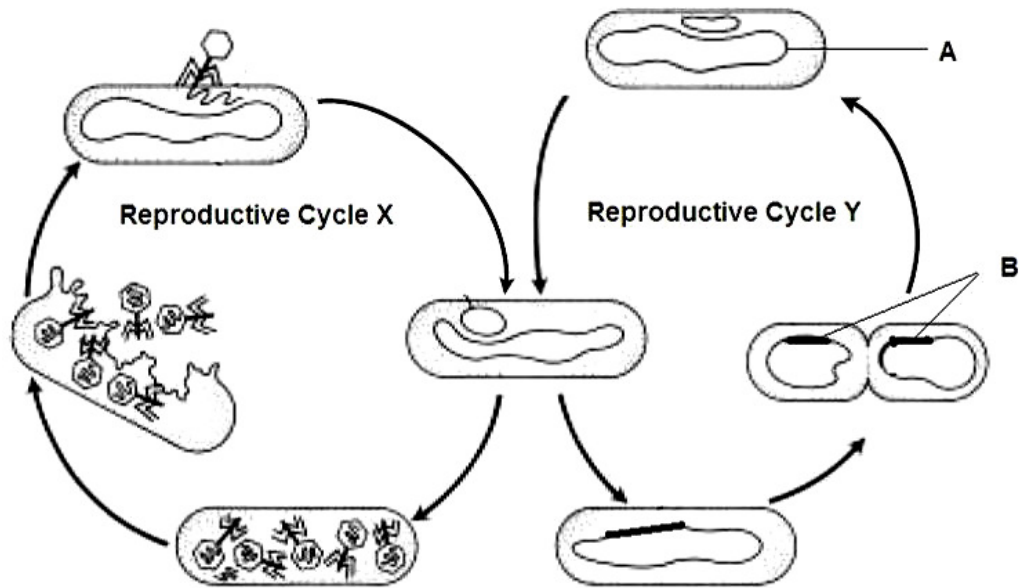
- 10 The following statements describe bacterial conjugation.

- I** The F plasmid is made of single-stranded DNA.
- II** When an F⁺ donor gives an F plasmid to an F⁻ recipient, both become F⁻.
- III** When an F⁺ donor gives an F plasmid to an F⁻ recipient, both become F⁺.
- IV** When an F⁺ donor gives an F plasmid to an F⁻ recipient, the donor becomes F⁻.
- V** When F⁺ cells are mixed with F⁻ cells, eventually all the cells will become F⁺.

Which of the statements are true?

- A** I, II and IV
- B** II and IV
- C** III, IV and V
- D** III and V

11 Which of the following correctly identifies the reproductive cycles **X** and **Y** and structures **A** and **B**?

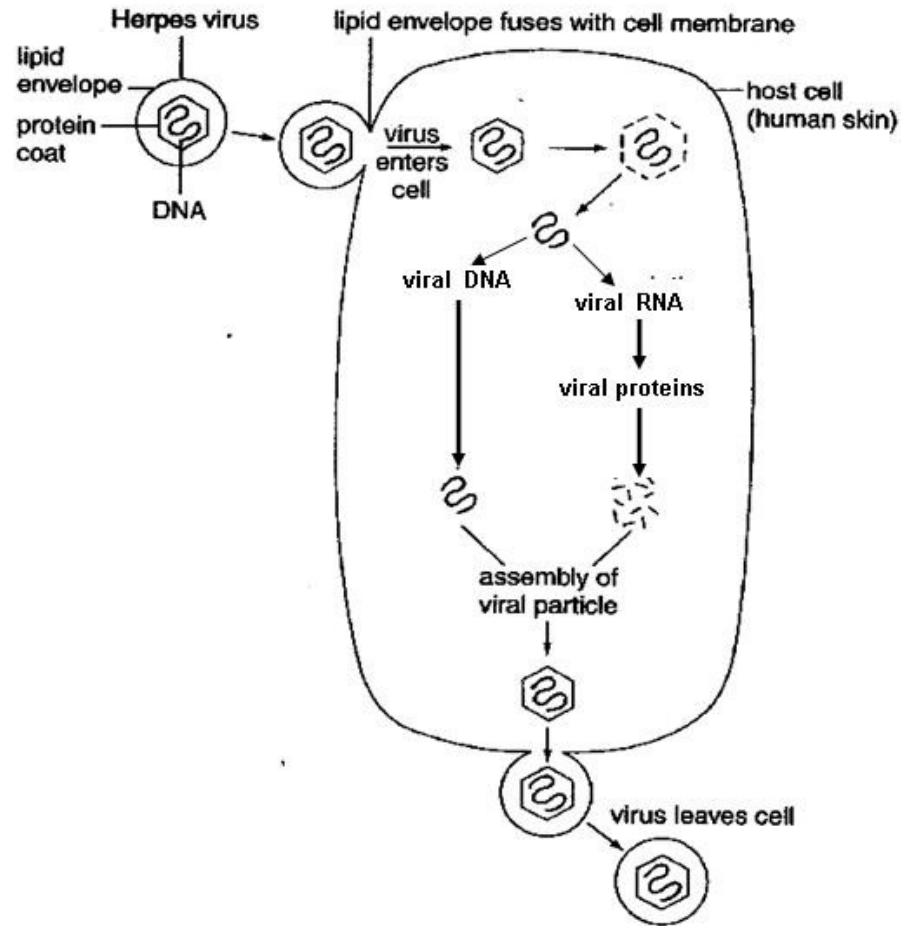


	X	Y	A	B
A	lytic	lysogenic	virus DNA	provirus
B	lysogenic	lytic	virus DNA	prophage
C	lytic	lysogenic	bacterial DNA	prophage
D	lysogenic	lytic	bacterial DNA	provirus

12 Which feature occurs in the life cycle of the influenza virus?

- A** Host cell DNA is destroyed by lytic enzymes.
- B** Viral genome is integrated into the host genome.
- C** Viral DNA acts as a template for DNA synthesis.
- D** Viruses enter the host cell by receptor-mediated endocytosis.

- 13 The diagram below shows the reproductive cycle of the herpes virus which causes cold sores on the mouth. With reference to the diagram below, which of the following statements best describes the herpes virus?

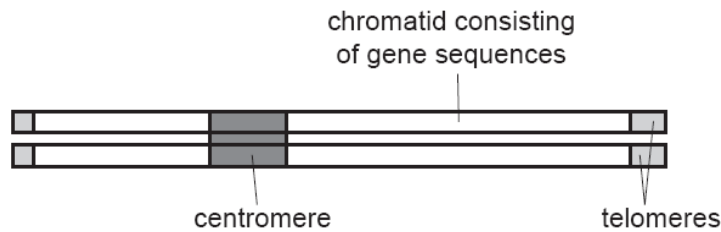


- A It is not a retrovirus as it does not contain RNA as its genetic material.
- B Its mode of replication is similar to that of HIV.
- C Its replication cycle includes a lysogenic phase
- D Death of the host cell is necessary for the release of the viral progeny.
- 14 Which of the following is NOT a feature of eukaryotic gene expression?
- A Polycistronic mRNAs are very rare.
- B Many genes are interrupted by noncoding DNA sequences.
- C RNA synthesis and protein synthesis are coupled.
- D mRNA is often extensively modified before translation.

- 15 Human telomeres consist of repeating TTAGGG sequences which extend from the ends of the chromosomal DNA.

When cells undergo mitotic division, some of these repeating sequences are lost. This results in a shortening of the telomeric DNA.

The diagram shows a eukaryotic chromosome.



What is a consequence of the loss of repeating DNA sequences from the telomeres?

- A The cell will begin the synthesis of different proteins.
 - B The cell will begin to differentiate as a result of the altered DNA.
 - C The number of mitotic divisions the cell can make will be limited.
 - D The production of mRNA will be reduced.
- 16 The acetylation of histones promotes loose chromatin structure. Recent evidence has shown that chemical modification of histones play a direct role in regulation of gene expression.

Which of the following best explains how acetylation regulates gene expression?

- A Helicase action is enhanced by acetylation.
 - B Acetylation of histones neutralizes their negative charges and encourages binding to DNA polymerase.
 - C RNA polymerase works better by binding with acetyl groups.
 - D When nucleosomes are highly acetylated, chromatin becomes less compact and DNA is more accessible for transcription.
- 17 The translation mixture contains a polynucleotide that directs the synthesis of Met-Gly-Gly-Phe-Leu-Ala. In the presence of Azithromycin, this polymer directs the synthesis of Met-Gly only.

From the information given, which of the following deductions could you make about Azithromycin?

	Control Stage	Conclusion
A	Translational	It prevents formation of the initiation complex, which contains the initiator tRNA and both ribosomal subunits.
B	Post translational	It inhibits binding of aminoacyl- tRNAs to the A site in the ribosome.
C	Translational	It blocks translocation of peptidyl transferase-rRNA from the A site to the P site of the ribosome.
D	Post translational	It interferes with chain termination and release of the peptide.

18 Which of the following genetic changes might lead to lower risk of cancer?

- 1** amplification of a tumour suppressor gene
- 2** deletion of the promoter upstream of a proto-oncogene
- 3** loss of function mutation of a proto-oncogene
- 4** translocation of a tumour suppressor gene to downstream of a strong promoter

- A** 1 and 3 only
- B** 2 and 3 only
- C** 1, 2 and 4 only
- D** 1, 2, 3 and 4

19 What do chromosomal aberrations and gene mutations have in common and how are they different?

	Similarity	Difference
A	Both may involve addition of nucleotides.	Gene mutations always produce dominant alleles but not chromosomal aberrations.
B	Both may not result in disorders.	Gene mutations do not involve inversions but inversions of segments of chromosomes do occur.
C	Both affect nucleotide sequence in DNA molecule.	Gene mutations occur within a chromosome but chromosomal aberrations may occur across chromosomes.
D	Both may not result in a difference in protein expression.	Chromosomal aberrations are more harmful than gene mutations.

20 In poultry, males have two X chromosomes and females have one X chromosome and one Y chromosome. The gene for feather-barring is sex-linked. The allele for barred feathers is dominant to the allele for non-barred feathers. A non-barred male is crossed with a barred female.

What ratio of offspring would be expected?

- A** 1 barred male : 1 barred female
- B** 1 non-barred male : 1 non-barred female
- C** 1 barred male : 1 non-barred female
- D** 1 non-barred male : 1 barred female

- 21 Purple buds of the morning glory flower, *Ipomoea*, open into blue flowers. As the flower opens, the pH on the vacuoles of the flower epidermal cells increases and this results in a change of colour from purple to blue.

A mutant purple-flowered morning glory plant carries recessive alleles of a gene **B/b**, coding for a membrane-bound ion pump, and is unable to increase the pH of the vacuole. Both normal blue flowers and mutant purple flowers have the same anthocyanin pigment, coded by the dominant allele of the gene **A/a**. Plants with **aa** cannot produce anthocyanin and they have white flowers.

The genes **A/a** and **B/b** are not linked.

A blue-flowered morning glory plant was crossed with a purple-flowered plant. Their offspring consisted of plants which are blue-flowered, purple-flowered as well as white-flowered.

What were the genotypes of the blue-flowered and purple-flowered plants?

	<i>Blue-flowered plant</i>	<i>Purple-flowered plant</i>
A	AABB	AaBb
B	AaBb	Aabb
C	aaBB	AaBB
D	aaBb	aaBb

- 22 The table below shows the blood group phenotypes resulting from crosses between different genotypes.

Genotype of 1 st parent	Genotype of 2 nd parent	
	I ^A I ^A	I ^A I ^O
I ^A I ^O	A	A and O
I ^B I ^O	A and AB	A, B, B and AB

Two parents have a son who has blood group O and phenylketonuria. One parent has blood group O and the other has blood group A. Neither parent has phenylketonuria.

What is the probability that the second child of these parents will be a girl with blood group A who does not have phenylketonuria?

- A** 1 in 16
B 1 in 8
C 3 in 16
D 3 in 8

- 23 When a pure-breeding maize plant with yellow seeds and full endosperm is crossed with another pure-breeding plant having white seeds and shrunken endosperm, the F_1 maize plants are found with yellow seeds and full endosperm. When the F_1 plants are selfed, four classes of descendants are obtained as shown in the table below:

Colour of seed	Shape of endosperm	Number of seeds observed in F_2 generation (O)
Yellow	Full	128
Yellow	Shrunken	27
White	Full	20
White	Shrunken	17

The formula for the chi-squared (χ^2) test is given as follows:

$$\chi^2 = \sum \frac{(O - E)^2}{E} \quad \text{where } \Sigma = \text{'sum of'}$$

O = observed 'value'
E = expected 'value'

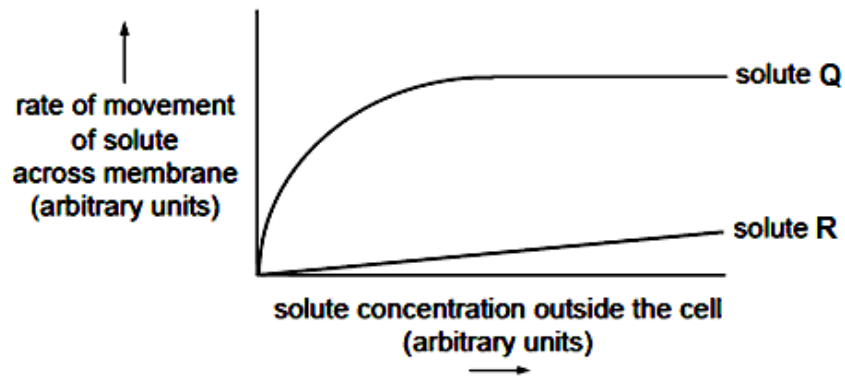
Degrees of freedom	Probability			
	0.10	0.05	0.01	0.001
1	2.71	3.84	6.64	10.83
2	4.69	5.99	9.21	13.82
3	6.25	7.82	11.35	16.27
4	7.78	9.49	13.28	18.47

Which of the following conclusions made about the inheritance of colour of seed and shape of endosperm is **correct**?

- A** Since $p < 0.05$, the difference between the observed and expected results is not significant. Thus the inheritance of colour of seed and shape of endosperm is following Mendel's law of independent assortment.
- B** Since $p > 0.05$, the difference between the observed and expected results is not significant. Thus the inheritance of colour of seed and shape of endosperm is not following Mendel's law of independent assortment.
- C** Since $p < 0.01$, the difference between the observed and expected results is not significant. Thus the inheritance of colour of seed and shape of endosperm is following Mendel's law of independent assortment.
- D** Since $p < 0.01$, the difference between the observed and expected results is significant. Thus the inheritance of colour of seed and shape of endosperm is not following Mendel's law of independent assortment.

- 24** Solute **Q** moves across the plasma membrane via protein channels. Solute **R** moves across the plasma membrane by simple diffusion. The rate of movement of each solute into a cell is recorded and graphed.

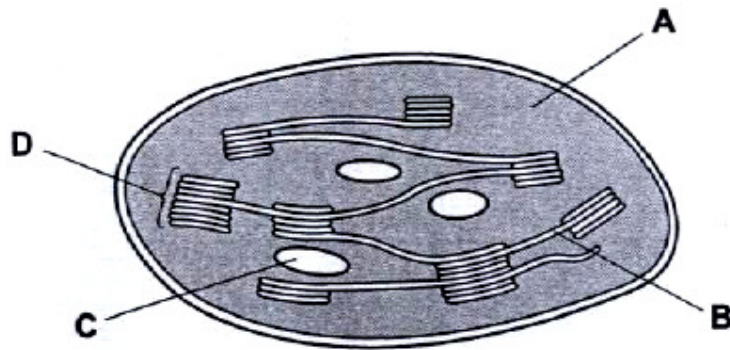
The results are shown in the following graph.



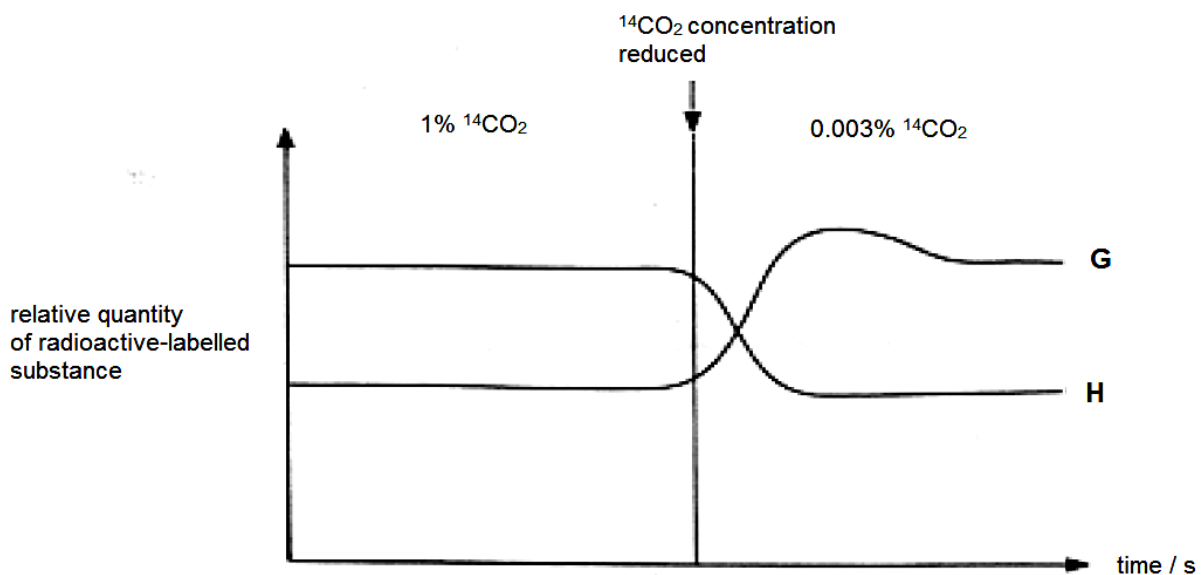
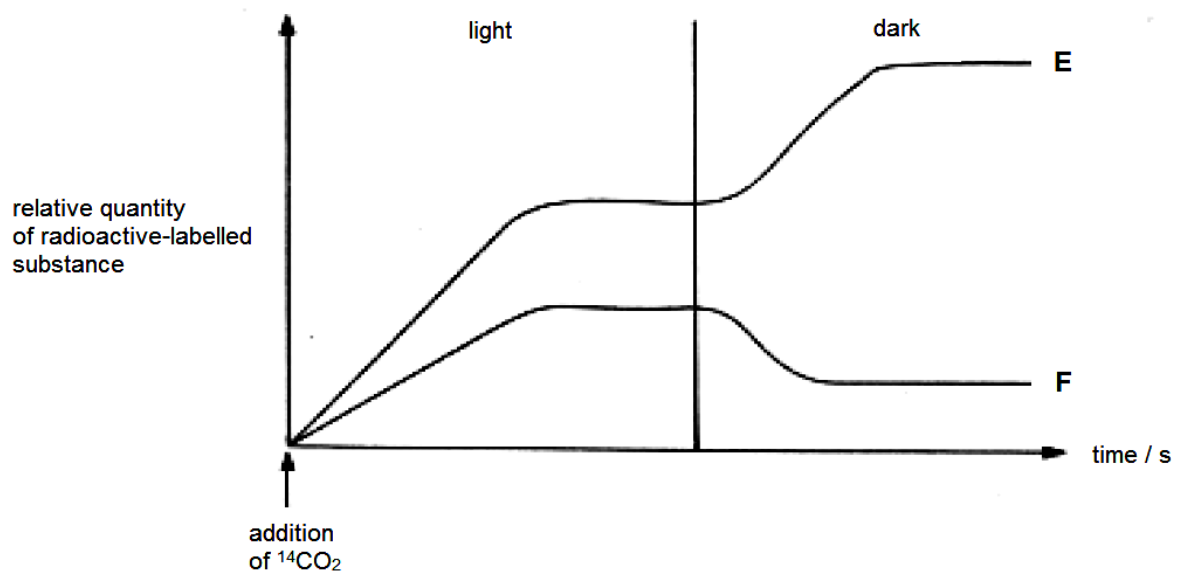
It would be reasonable to conclude that

- A** solute **Q** is lipid soluble.
 - B** solute **Q** saturates the protein channels.
 - C** the gradient of the graph for solute **R** will increase as the temperature decreases.
 - D** as solute **R** is metabolised within a cell, its rate of movement across the membrane decreases.
- 25** The diagram shows a section through a chloroplast.

Where would the products of photophosphorylation be used?



- 26 The following graphs shows two separate experiments on carbon dioxide fixation in photosynthesis using a unicellular green alga and radioactive isotope ^{14}C to label the carbon dioxide. The algae were actively photosynthesizing before the start of both experiments.



Which of the following correctly identify the graphs **E**, **F**, **G** and **H**?

	E	F	G	H
A	RuBP	GP	GP	RuBP
B	GP	RuBP	GP	RuBP
C	GP	RuBP	RuBP	GP
D	RuBP	GP	RuBP	GP

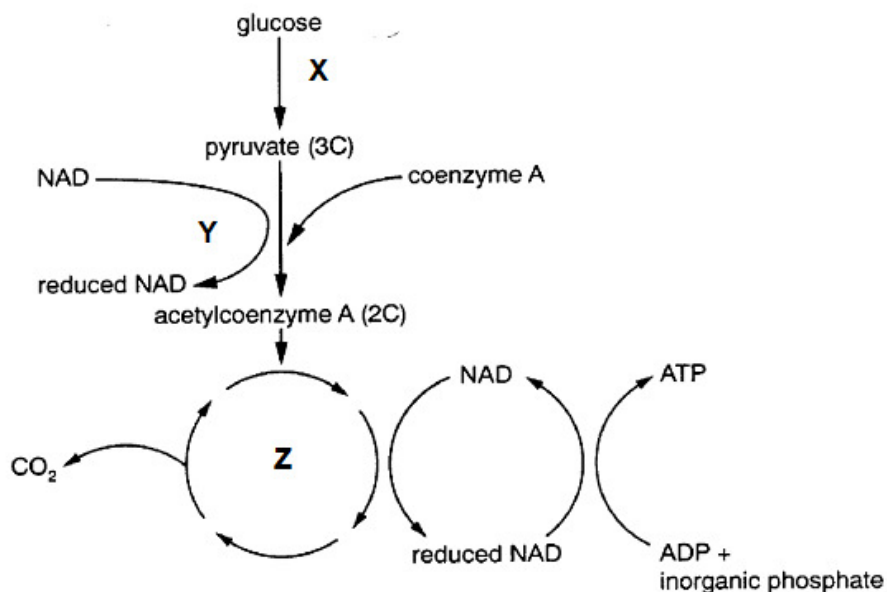
- 27 Rotene and oligomycin are two metabolic poisons which affect cellular respiration. The effects of rotene and oligomycin on aerobic respiration are summarised in the table.

	Ability to use glucose	Ability to use oxygen	ATP yield
Rotene	Yes	No	Decreases
Oligomycin	Yes	Yes	Decreases

Which of the following correctly identifies the specific functions of these two metabolic poisons?

- | | Rotene | Oligomycin |
|----------|------------------------------|------------------------------|
| A | Electron transport inhibitor | Inhibits ATP synthase |
| B | Inhibits ATP synthase | Electron transport inhibitor |
| C | Dissipate proton gradient | Inhibits ATP synthase |
| D | Inhibits ATP synthase | Dissipate proton gradient |

- 28 The flow chart shows a series of reactions occurring in an animal cell.



Which of the following statements correctly describes the flow chart?

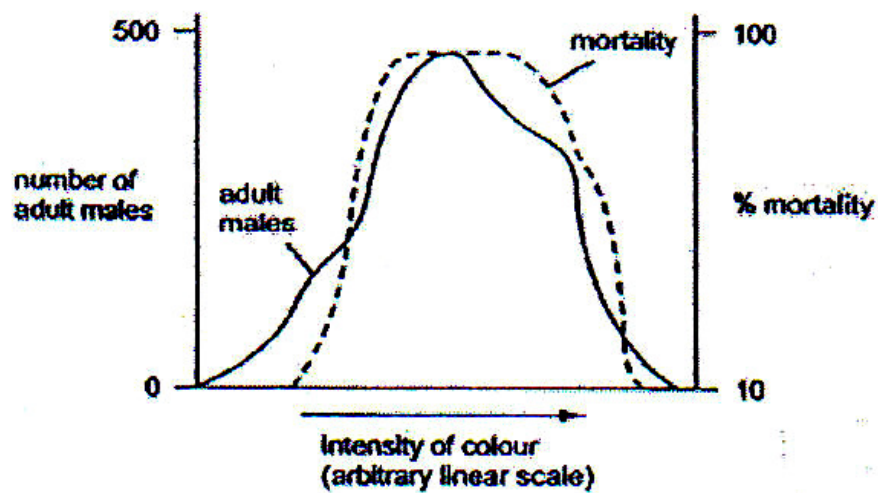
- A** Reaction **X**, which occurs in the cytosol, is an anabolic reaction.
- B** Reaction **Y** involves the process of substrate-level phosphorylation, whereby pyruvate is first converted to a compound called acetyl coenzyme A.
- C** Reaction **Y** occurs in the cytoplasm whereas reaction **Z** occurs in the mitochondria.
- D** Reaction **Z** is a catabolic pathway which occurs twice for every glucose molecule to be completely oxidised.

29 Which sequence of events correctly describes evolution?

- 1 Differential reproduction of the spiders occurs.
- 2 A new selection pressure occurs.
- 3 Allele frequencies within the spider population change.
- 4 Poorly adapted spiders have decreased survivorship.

- A 2 4 1 3
B 2 4 3 1
C 4 1 3 2
D 4 3 1 2

30 The graph below shows data on a population of a species of moth which shows considerable variation in colour intensity



Which conclusion can be made from this graph?

- A Colour variation is environmentally induced.
B Colour variation is genetically determined.
C Extreme forms are favoured by natural selection.
D The species shows discontinuous variation with respect to colour.

- 31** Darwin's view of the process of evolution to form new species (speciation) has been reinforced by more recent discoveries in genetics and cell biology. This is termed the neo-Darwinian view of evolution.

In this view, which sequence of events is considered most likely to lead to speciation?

A	adaptation of population	→	competition and predation leading to natural selection	→	behavioural isolation	→	sympatric speciation
B	adaptation of population	→	competition and predation leading to natural selection	→	behavioural isolation	→	allopatric speciation
C	competition and predation leading to natural selection	→	geographical isolation	→	adaptation of isolated populations	→	sympatric speciation
D	competition and predation leading to natural selection	→	geographical isolation	→	adaptation of isolated populations	→	allopatric speciation

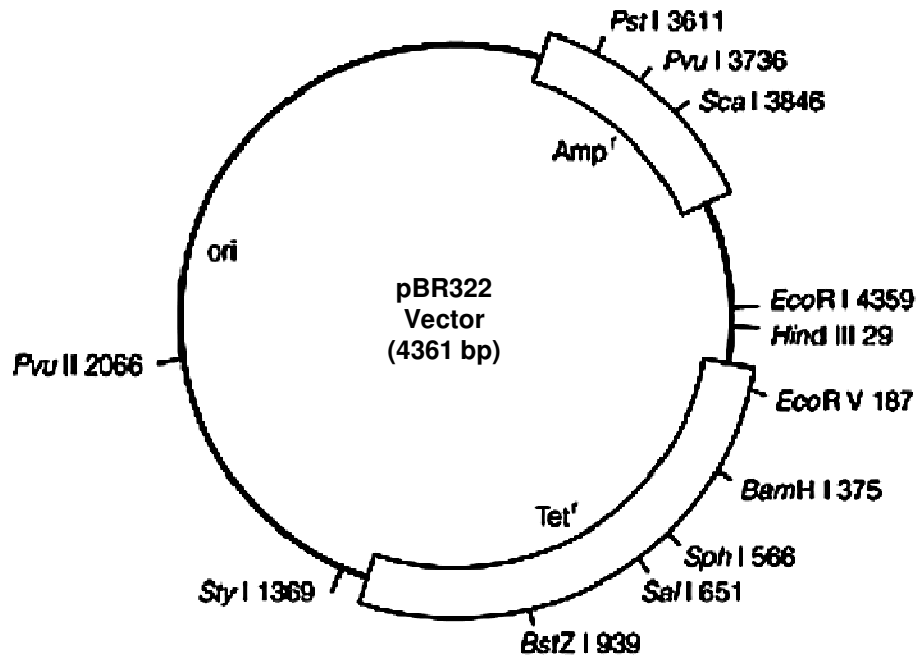
- 32** The table shows amino acid sequences from a segment of a viral protein that is found in a number of different viruses. These viruses were isolated from patients suffering from either the common cold caused by rhinovirus or flu caused by influenza. Each letter represents a single amino acid.

Virus	Amino acid sequence																	
Rhinovirus 1	P	E	F	P	Y	N	A	T	T	E	P	T	K	A	V	P	F	Q
Rhinovirus 2	P	E	F	S	Y	S	A	V	D	D	P	I	G	E	E	P	F	K
Rhinovirus 3	P	E	F	S	Y	S	A	G	D	D	P	A	G	E	E	P	F	N
Influenza A	P	E	Y	H	T	P	V	M	E	P	L	D	P	F	T	M	D	K
Influenza B	P	E	Y	H	T	P	K	M	E	P	L	D	A	F	A	M	D	N
Influenza C	P	E	Y	H	T	P	V	M	E	P	L	D	V	F	D	M	D	K

Which of the following statements is consistent with the data given?

- A** The rhinoviruses are more closely related to each other than the influenza viruses are to each other.
- B** Influenza virus **A** is more closely related to influenza virus **B** than influenza virus **B** is to influenza virus **C**.
- C** The influenza viruses are less closely related to each other than they are to the rhinoviruses.
- D** Rhinovirus **2** is more closely related to rhinovirus **3** than it is to rhinovirus **1**.

- 33 The pBR322 vector is used to clone a eukaryotic gene, which has been digested by the restriction endonuclease *Bam*HI.



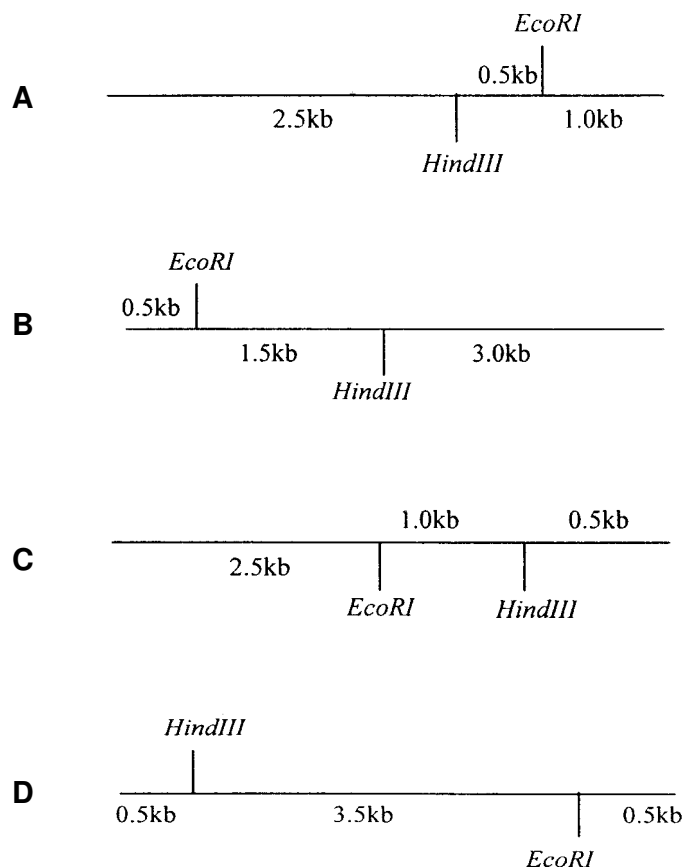
Following transformation, bacterial cells were grown in four different media, as shown below:

- 1 Nutrient broth containing ampicillin
- 2 Nutrient broth containing tetracycline
- 3 Nutrient broth containing ampicillin and tetracycline
- 4 Nutrient broth without ampicillin and tetracycline

Which of the following media would bacterial cells containing the recombinant plasmids grow in?

- A 4
- B 1 and 2
- C 2 and 3
- D 1 and 4

- 34 Digestion of a 4 kb DNA molecule with *EcoRI* yields two fragments of 1 kb and 3 kb each. Digestion of the same molecule with *HindIII* yields fragments of 1.5 kb and 2.5 kb. Finally, digestion with *EcoRI* and *HindIII* in combination yields fragments of 0.5 kb, 1 kb and 2.5 kb. How would a restriction map indicating the positions of the *EcoRI* and *HindIII* cleavage sites look like?



- 35 The dashed lines in the template sequence represent a long sequence of bases to be amplified.

Template

5' ATTCGGACTTG ----- GTCCAGCTAGAGG 3'
 3' TAAGCCTGAAC ----- CAGGTCGATCTCC 5'

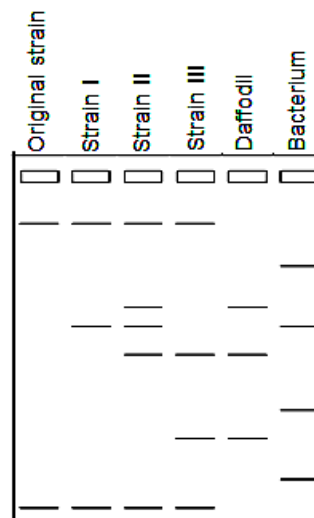
Which of the following sets of primers can be used in the PCR for the amplification of the following DNA sequence?

- A** 5' GTCCAGC 3' & 5' CCTGAAC 3'
B 5' ATTCGGA 3' & 5' CCTCTAG 3'
C 5' GGA CTTG 3' & 5' GCTGGAC 3'
D 5' AUUCGGA 3' & 5' GAUCUCC 3'

- 36** Which of the following statements about the human genome project (HGP) is **false**?
- A** HGP aims to identify all the genes in human and to determine the DNA sequences of these genes.
 - B** HGP aims to allow genetic testing to take place for earlier detection of genetic diseases
 - C** HGP allows defective genes to be replaced through gene therapy
 - D** HGP allows comparative studies to be made between humans and other organisms to identify similar genes associated with diseases.
- 37** What is the role of stem cells with regard to the function of adult tissues and organs?
- A** Stem cells are undifferentiated cells that divide asymmetrically, giving rise to one daughter cell that remains a stem cell and one daughter that will differentiate to replace damaged and worn out cells in the adult tissue or organ.
 - B** Stem cells are embryonic cells that persist in the adult, and can give rise to all of the cell types in the body.
 - C** Stem cells are differentiated cells that have yet to express the genes and produce proteins characteristic of their differentiated state, and do so when needed for repair of tissues and organs.
 - D** Stem cells are fully differentiated cells that reside under the surface of epithelial tissue, in position to take over the function of the tissue when the overlying cells become damaged or worn out.
- 38** Which of the following regarding embryonic stem cells and hematopoietic stem cells is true?
- A** As embryonic stem cells develop, they turned into hematopoietic stem cells as they lose their ability to differentiate into all types of cells.
 - B** Embryonic stem cells have more genes than hematopoietic stem cells and thus are able to form more types of cells.
 - C** Under normal conditions, embryonic stem cells express more of their genes compared to the hematopoietic stem cells.
 - D** Both stem cells are derived from the zygotic stem cells with the hematopoietic stem cells having a lowered telomerase activity compared to the embryonic stem cells.
- 39** Some scientists are concerned about the release of genetically modified microorganisms into their natural habitat.
- What is the most likely reason for this concern?
- A** The microorganisms may reproduce quickly.
 - B** The microorganisms may not survive.
 - C** The mutation rate of the microorganisms would increase.
 - D** The transfer of changed genes to other organisms.

- 40** An attempt was made to produce Golden rice. To determine whether or not DNA from the daffodils and the bacterium had been successfully incorporated in the DNA of the rice, scientists used PCR and gel electrophoresis to produce DNA profiles.

The following DNA profiles belong to the original strain of rice, three strains **I** to **III** of genetically modified Golden rice, and the species of daffodil and bacterium used to incorporate beta-carotene genes in the rice.



Which one of the strain(s) of Golden rice has successfully incorporated DNA from both the daffodil and the bacterium?

- A** Strain **I** only
- B** Strain **II** only
- C** Strain **I** and **III** only
- D** Strain **II** and **III** only

END OF PAPER

2013 CJC H2 Biology Prelim Paper 1 Answers

1	D	21	B
2	B	22	C
3	B	23	D
4	A	24	B
5	C	25	A
6	C	26	C
7	D	27	A
8	B	28	D
9	C	29	A
10	D	30	C
11	C	31	D
12	D	32	D
13	A	33	D
14	C	34	A
15	C	35	B
16	D	36	B
17	C	37	A
18	D	38	D
19	C	39	D
20	C	40	B